

● *Original Contribution***THE DISAPPEARANCE OF ULTRASOUND CONTRAST BUBBLES:
OBSERVATIONS OF BUBBLE DISSOLUTION AND
CAVITATION NUCLEATION**

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Abstract—The destruction process of biSphere™ and Optison™ ultrasound (US) contrast microbubbles were studied at 1.1 MHz. High-amplitude tone bursts caused shell disruption and/or fragmentation of the microbubbles, leading to dissolution of the freed gas. The bubble destruction and subsequent dissolution process was imaged with a high pulse-repetition frequency (PRF) 10-cycle, 5-MHz bistatic transducer configuration. Three types of dissolution profiles were measured: In one case, biSphere™ microbubbles showed evidence of dissolution through resonance, during which a temporary increase in the scattering amplitude was observed. In another case, both biSphere™ and Optison™ microbubbles showed evidence of fragmentation, during which the scattering amplitude decreased rapidly. Finally, in some cases, we observed the impulsive growth and subsequent rapid decay of signals that appear to be due to cavitation nucleation. Simulations of bubble dissolution curves show good agreement with experiments. (E-mail: matula@apl.washington.edu) © 2002 World Federation for Ultrasound in Medicine & Biology.

Key Words: Ultrasound, Dissolution, Cavitation, Inertial cavitation, Nucleation, Ultrasound contrast agents, Microbubbles.

INTRODUCTION

Ultrasound (US) contrast agents (essentially stabilized microbubbles) are currently being refined to enhance the sensitivity and specificity of diagnostic US imaging (Goldberg 1997). The agents have a typical size distribution centered around 3 to 4 μm in diameter. They are stabilized against dissolution by a coating that may consist of an albumin, polymer or lipid shell, filled with various gases. Imaging applications include, but are not limited to (Sieswerda et al. 2000; Lombardo and Loperfido 2001), 1. the status of myocardial perfusion and the location of myocardial infarction, 2. delineation of the endocardial borders in certain technically difficult areas during echocardiographic examinations, 3. evaluation of the left ventricular wall motion and function, and 4. quantitative assessment of the coronary blood flow.

Some imaging techniques rely on contrast bubble destruction. For example, transient enhancement of

power Doppler intensities and noise-like ‘pseudo-Doppler’ shifts were detected in tissue preperfused with US contrast agents after insonification with high-intensity acoustic pulses (stimulated acoustic emission) (Blomley et al. 1999; Pohl et al. 2000; Tiemann et al. 2000). Also, subtraction of the B-mode images acquired before and after the high-intensity pulses (flash echo) increases the sensitivity of detection and eliminates the effect of tissue motion (Pelberg et al. 1999; Frinking et al. 2001). High-intensity short-duration pulses followed by low-amplitude ones allow the assessment of the perfusion status of the myocardium and other tissues by replenishment of contrast microbubbles (Wei et al. 1998; Schlosser et al. 2001). These techniques apparently depend on microbubble destruction (Kamiyama et al. 1999; Tiemann et al. 2000; Porter et al. 2001).

In addition to imaging, the destruction of contrast microbubbles is also being considered for therapeutic applications. The rapid expansion and violent collapse of gas bubbles, leading to inertial cavitation (IC), is believed to be an important mechanism for certain beneficial US-induced bioeffects within the body, such as hemostasis (Vaezy et al. 2001), drug delivery (Shohet et

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al. 2000), and thrombolysis (Everbach and Francis 2000). For example, site-specific drug delivery can be achieved by destroying drug-filled contrast microbubbles in the target area with high-intensity US (Porter et al. 1996). Shohet et al. (2000) found that albumin-coated microbubbles could be used to effectively deliver an adenoviral transgene to rat myocardium via US-mediated microbubble destruction. Furthermore, introduction of US contrast agents together with US at the thrombosis section of blood vessels was found to increase the dissolving rate of clots (Nishioka et al. 1997). It is, therefore, important to understand the mechanisms of bubble destruction, which can also result in nucleation, as well as how to control bubble behavior using various acoustic parameters.

In previous research, shelled microbubbles were found to rupture and release their gas content beyond a certain pressure threshold. Researchers successfully showed the sudden increase of scattering signals from Quantison™ microbubbles after insonifying with a low-frequency, high-amplitude US burst (Bouakaz et al. 1999). The “enhanced scattering” effect was generated by the released gas bubbles with higher scattering ability. These free gas bubbles subsequently disappeared due to dissolution. Shi et al. (2000), using Sonazoid™ microbubbles, also showed that, after a higher-amplitude US burst, shelled microbubbles probably lost their shells and dissolved in the solution quickly. Ferrara’s group, using a high-speed camera, successfully showed the slow diffusion of gas through the shell and the formation of a shell defect followed by rapid diffusion of gas into the surrounding liquid (Dayton et al. 1999). The low-solubility gas can persist for several hundred ms in solution. Their studies on the mechanisms of contrast agent destruction tell us that fragmentation, static diffusion and acoustically driven diffusion are possible mechanisms of contrast microbubble destruction (Chomas et al. 2001).

Following the work of Bouakaz and Shi, we have performed studies on the destruction of contrast microbubbles. The objective of our research was to better understand the pulse-echo data, such as that shown in Figs. 1 and 2. These figures show evidence of the appearance of nucleated microbubbles (the sudden appearance of echoes) accompanied by the apparent destruction of contrast microbubbles (the disappearance of echoes). Although pulse-echo data provide relatively high-resolution information to differentiate echoes, we believed that, by employing longer interrogating tone bursts, we could obtain even more information. Using these longer tone bursts, our goals were to 1. measure the dissolution process of shell-disrupted microbubbles, 2. use the resonance features of dissolving bubbles to discriminate between bubbles of various sizes, 3. detect and observe cavitation nucleation, and 4. compare the experimental

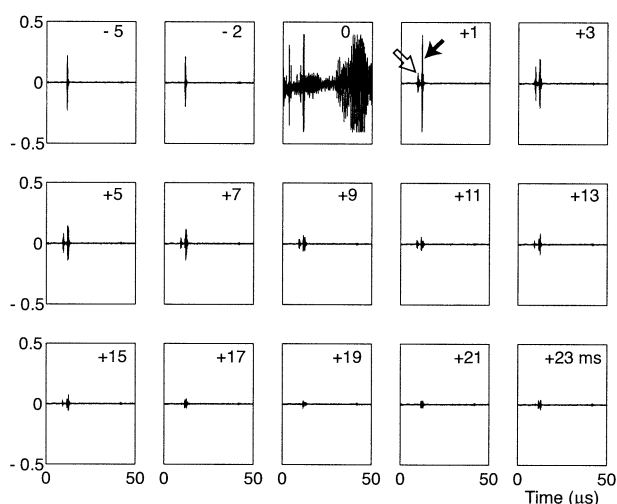


Fig. 1. Selected echo segments from biSphere™ scattering data before and after a high-intensity destructive acoustic (HIFU) pulse at segment 0 ms. After the HIFU pulse, the original bubbles began to undergo a series of amplitude changes (solid arrow). A new echo also appeared immediately after the HIFU pulse, a few μ s before the primary echo (hollow arrow). The new signal is apparently from fragmented or nucleated microbubbles.

data with mathematical simulations of bubble dissolution.

Theory

Our goal in this section was to simulate the scattering of sound from bubbles that have had their shells disrupted and are, thus, free to dissolve. We begin with a description of the dissolution of a single free gas bubble, taking into account that the bubble may contain gases other than air. For example, Optison™ agents contain octofluoropentane (OFP). Because our experiments involved the scattering from a distribution of bubbles, we also included the effects of a small size distribution. Finally, we incorporated the changing radius of the dissolving bubbles $R(t)$ into the scattering equations to obtain a scattering cross-section vs. time simulation that we could then compare with experiments.

Simulation of a free bubble dissolution. We considered the dissolution process of a shell-disrupted free gas bubble. This assumption is not justified *a priori*, because one can envision bubbles attached to shell fragments or relatively intact shelled bubbles with cracks in the shell. The dissolution rate of these partially intact bubbles, which depends on the surface area of the bubble exposed to the surrounding medium, would be expected to be decreased. However, by comparing the simulated dissolution from free gas bubbles with the experimental data, we hoped to reasonably elucidate the destruction process.

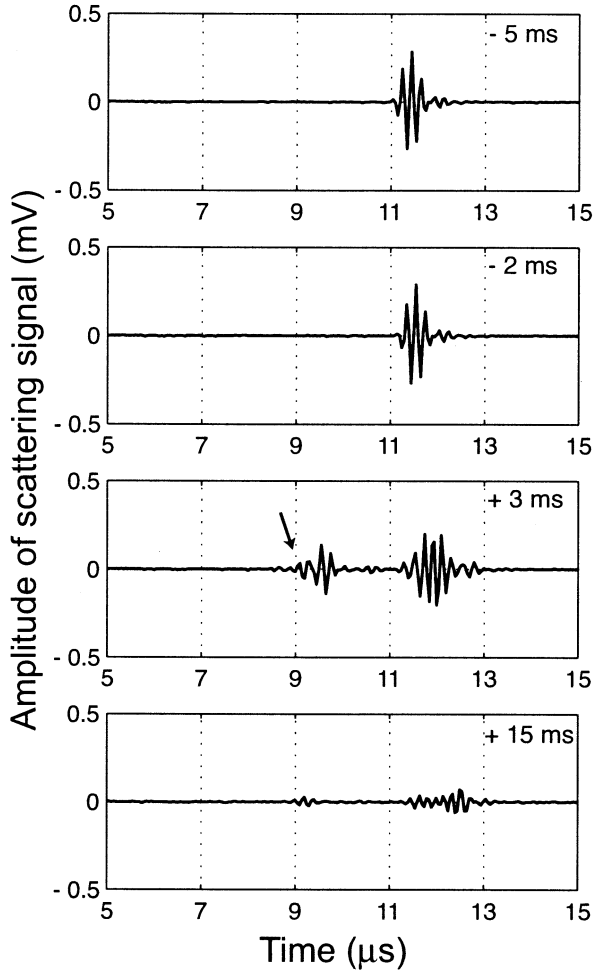


Fig. 2. Closeup of the echoes of 4 recorded segments from the data associated with Fig. 1. The time marker is relative to the HIFU disruption pulse. All signals before the HIFU pulse show only a minor change in arrival time, suggesting that its motion is nearly stationary. After the HIFU pulse, a new signal appears (arrow). Subsequently, both signals decay; again, with little motion. The echo decay is presumably due to bubble dissolution.

For an air bubble in water, the dissolution rate is dependent on its initial size R_0 , as well as the concentration of air in solution. The dissolution rate can be calculated by the following equation (Epstein and Plesset 1950):

$$\frac{dR}{dt} = -D_A L_A \frac{1 - \frac{c_\infty}{c_0} + \frac{\tau}{R\rho_g}}{1 + \frac{2\tau}{3R\rho_g}} \left(\frac{1}{R} + \frac{1}{\sqrt{\pi D_A t}} \right), \quad (1)$$

$$\tau = \frac{2M\sigma}{R_G T}$$

All parameters are defined in Table 1. The initial condition used in the calculation is $R(t = 0) = R_0$.

Equation (1) describes the dissolution of a bubble in a single-component system; that is, the gas dissolved in the fluid is the same as the gas within the bubble. With agents such as Optison™, that contain a less water-soluble gas such as octofluoropentane (OFP), we must consider a multicomponent system in which a bubble may be filled with one gas (such as OFP) within a solution of a different dissolved gas (such as air). To model the dissolution process more accurately, we must incorporate two independent gas diffusion and dissolution processes. Our calculation is based on equations (2), (3) and (5), two diffusion equations and a pressure balance equation. (Kabalnov *et al.* 1998).

$$\frac{d}{dt} (R^3 C_A) = -3R^2 D_A \left(C_A - \frac{c_\infty}{c_0} \frac{P_0}{R_G T} \right) L_A \left(\frac{1}{R} + \frac{1}{\sqrt{D_A t}} \right) \quad (2)$$

$$\frac{d}{dt} (R^3 C_F) = -3R^2 D_F C_F L_F \left(\frac{1}{R} + \frac{1}{\sqrt{\pi D_F t}} \right), \quad (3)$$

where we kept the $1/\sqrt{\pi D t}$ term in our derivation, although it is usually sufficient to neglect it because it is much smaller than the $1/R$ term. For example, consider the case of a 3.2- μm diameter bubble filled with a 50% mixture of OFP and air. The dissolution times with and without the $\pi D t$ term are 2.30 s and 2.37 s, respectively.

Equation (2) gives the dissolution rate for the bubble in terms of the air concentration, and eqn (3) gives the dissolution rate in terms of the OFP gas concentration [note that eqn (2) reduces to eqn (1) for a single-component system]. Simultaneous solution of eqns (2) and (3) leads to determining the dissolution of a bubble in a mixed-gas environment. The initial conditions are given by $R(t = 0) = R_0$ and the volume fraction:

$$X_F = \frac{C_F(t = 0)}{C_A(t = 0) + C_F(t = 0)}. \quad (4)$$

Solution of the equations is obtained with the help of the ideal gas equation of state, $P_{A,F} = C_{A,F} R_G T$ and, realizing that a pressure balance exists between the partial pressures of the gases and the external ambient pressure, including the surface tension pressure:

$$p_A + p_F = \frac{2\sigma}{R} + P_0. \quad (5)$$

After we know the size of a bubble at any time, we can then calculate the scattering cross-section as a function

Table 1. Notation used throughout paper

Symbol	Description	Value
C_A	Molar concentration of air inside the bubble (mol/m ³)	
C_F	Molar concentration of C ₃ F ₈ inside bubble (mol/m ³)	
c	Sound speed at 25°C (m/s)	1500*
c_∞	Concentration of dissolved gas in solution (kg/m ³)	
c_0	Saturation concentration of dissolved gas (kg/m ³)	
D	Diffusion coefficient (m ² /s)	
D_A	Diffusion coefficient of air (m ² /s) at 25°C	2×10^{-9} *
D_F	Diffusion coefficient of C ₃ F ₈ (m ² /s)	7.45×10^{-10} †
f	Frequency (Hz)	
$H(\omega)$	Transfer function of 5-MHz transducers	
L_A	Ostwald coefficient of air	1.71×10^{-2}
L_F	Ostwald coefficient for C ₃ F ₈	5.20×10^{-4} †
M	Molecular weight of gas (kg/mol)	
P_v	Vapor pressure of perfluorocarbon (Pa)	
P_A	Air pressure inside the bubble (Pa)	
P_F	Perfluorocarbon gas pressure inside the bubble (Pa)	
P_0	Ambient pressure (Pa)	10^5 *
R_G	Gas constant (J/mol°K)	8.314*
R	Radius of bubble (m)	
R_0	Initial radius of bubble (m)	
T	Temperature (°K)	298
t	Time (s)	
V_m	Molar volume (cm ³ /mol)	
X_F	Volume fraction of C ₃ F ₈ in the bubble (= 1 for 100%)	
σ	Surface tension (N/m)	73×10^{-3} *
σ_s	Scattering cross-section (m ²)	
ρ_w	Density of water (kg/m ³)	1000*
ρ_g	Density of air (kg/m ³)	1.16*
δ	Total damping factor	
δ_{th}	Thermal damping factor	
δ_η	Viscous damping factor	
δ_R	Radiation damping factor	
η_L	Shear viscosity of water (kg/m·s)	0.001*
Ω	Normalized frequency (ω/ω_0)	
ω	Frequency (rads)	
ω_0	Resonance frequency (rads)	

* The CRC Handbook of Chemistry and Physics, 80th edition; † See Appendix A.

of time. This will then be used to calculate the expected scattering amplitude vs. time that we would see on an oscilloscope. That is, within a scaling factor, the oscilloscope voltage signals will be the square root of the scattering cross-section.

The calculation of the scattering cross-section is based on the assumptions of scattering from small particles (the wavelength $\lambda \gg R$ or $kR \ll 1$; k is the wave number).

imental parameters, the thermal damping constant is an order of magnitude smaller than the sum of viscous and radiation damping constants. For example, the thermal damping factor $\delta_{th} = 0.032$ for a 2 μ m-radius bubble, and $\delta_\eta + \delta_R = 0.199$. For a fragmented microbubble of 0.5- μ m radius, the viscous damping factor dominates ($\delta_\eta = 0.233$ and $\delta_{th} = 0.015$). Therefore, for simplicity, we neglect the thermal damping factor. The viscous and radiation damping terms are given by:

$$\sigma_s = 4\pi R^2 \frac{\Omega^4}{(\Omega^2 - 1)^2 + \Omega^2 \delta^2}. \quad (6)$$

$$\delta_\eta = \frac{4\eta_L}{\omega_0 \rho_w R^2} \quad (7)$$

$$\delta_R = \frac{\omega^2 R}{\omega_0 c}. \quad (8)$$

where $\Omega = \omega/\omega_0$ and δ is the total damping constant. Although the damping is light, we must include it to keep the cross-section finite at the resonant frequency. For a free bubble, the constant includes thermal (δ_{th}), viscous (δ_η) and radiation damping (δ_R). However, for the range of sizes that exist for contrast agents, and for our exper-

See the references for the detailed description and derivations of these damping factors (Church 1995; Hoff

2000; Hoff *et al.* 2000). The total damping is thus $\delta \approx \delta_\eta + \delta_R$.

Multiple bubble assumption. In the previous subsection, we considered the scattering from a single bubble. In our experiments, we injected a volume of contrast agents containing many microbubbles into the solution; thus, we must consider that the focal zone contains a small size distribution of bubbles. For simplicity, we assume that the size distribution of shell-disrupted or nucleated bubbles $W(R_0)$ is a Gaussian function with mean radius of R_0 and SD, within a fixed range of radii ($rr = 0$ to $3 \mu\text{m}$ radius).

$$W(R_0) = \frac{1}{\sqrt{2\pi(SD)}} e^{-\frac{(rr - R_0)^2}{2(SD)^2}}. \quad (9)$$

MATERIALS AND METHODS

All experiments were performed in a $16.5 \times 12.5 \times 12.5$ cm acrylic tank containing deionized, filtered water. The contrast agents used in the experiments were biSphere™ (Point Biomedical, San Carlos, CA) and Optison™ (Amersham Health, Princeton, NJ) (Podell *et al.* 1999) microbubbles, diluted to the concentration of 1 to $5 \mu\text{L/L}$. The former is an experimental agent with air enclosed by an inner polymer and an outer albumin shell. The latter is commercially available, and has OFP gas enclosed with an albumin shell. The mean radius is $2 \mu\text{m}$ for biSphere™ and ranges between 1 and $2.25 \mu\text{m}$ for Optison™ (according to the label).

Destructive US pulses were achieved using a 1.1-MHz (7.0 cm diameter, 5.5 cm focal length) single element, high-intensity focused US (HIFU) transducer, mounted onto one side of the tank. Its measured -3 dB profile is approximately 1 mm wide by 14 mm long. Initially, a bistatic pulser-receiver detection system was used to measure bubble destruction and dissolution. This system consisted of two 5-MHz transducers (1.2 cm diameter, 4.6 cm focal length), mounted in quadrature on two of the remaining tank walls (see Fig. 3). One of the bistatic transducers interrogated the HIFU focal region with low-amplitude acoustic pulses, and the second transducer collected the scattering signals from the focal zone. The measured -3 -dB profile of these transducers is approximately 0.4×5 cm. A recent modification to the system incorporates a pulser-receiver TR switch that permits long tone bursts, allowing us to remove one of the 5 MHz transducers. We currently use this monostatic interrogator-receiver system, which gives similar results to the bistatic system.

Separate function generators (33120A, Hewlett Packard, Loveland, CO) were used to control the HIFU and interrogating transducers. A power amplifier (400B,

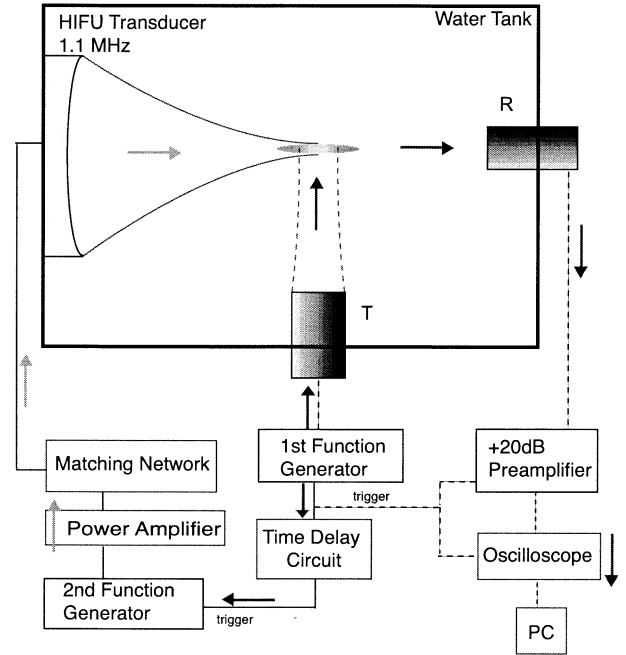


Fig. 3. HIFU transducer generates destructive acoustic pulse at focus. A bistatic 5-MHz transducer system interrogates (T) and receives (R) the scattering signal from bubbles in the focal region. All three transducers are focused at the same region. A trigger delay circuit is used to provide a variable time delay between the HIFU and interrogating pulses.

ENI Amplifier, Rochester, NY) set at the maximal leveled output with a constant gain of 74 dB powered the HIFU transducer. The pulse-repetition frequency (PRF) for the HIFU transducer was set at 2 to 10 Hz, depending on the experiment. Each HIFU pulse was followed by 500 to 2500 interrogating pulses (PRF = 5 kHz). A time delay circuit was used to synchronize each device and provide the necessary delay (Fig. 3).

The peak pressure levels used for shelled bubble destruction ranged from 1.5 to 4 MPa. These pressures are higher than the inertial cavitation (IC) threshold at 1.1 MHz (unpublished observations). The pressure level for the 5-MHz interrogating pulse was approximately 0.09 MPa, which was not able to induce IC nor cause destruction to the shelled microbubbles. This was verified by noting that the echo signal remained constant in amplitude and position for long periods of time, indicating that the bubble remained stationary in size and location.

Experiment protocol

Part 1: Contrast agent destruction and bubble dissolution. For all experiments, the water was left overnight to achieve an equilibrium gas saturation concentration. Bubbles attached to the transducers and chamber

walls were removed before the experiments. The experiment protocol then consisted of injecting contrast agents (1 to 5 $\mu\text{L/L}$) into the water and stirring the solution using a magnetic stirrer operating at low speed. The stirrer was switched off and the water was allowed to settle approximately 1 min before the experiment began.

Interrogating pulses (10 cycles, 5 MHz, PRF = 5 kHz) were then turned on to detect the shelled bubbles and to ensure that the bubbles remained relatively stationary over short time scales. The pulse length of the HIFU transducer was set to 5 cycles to minimize the possibility of streaming. For detecting the dissolution of the less soluble gas in Optison™ microbubbles, we used the same protocol, but decreased the HIFU PRF to 1 Hz, and the PRF of the interrogating pulse to 500 Hz, to capture longer dissolution times.

Part 2: Bubble nucleation and inertial cavitation.

In the absence of microbubble contrast agents, HIFU pulses did not generate scattering signals above background. However, at much higher amplitudes (> 4 MPa), HIFU pulses were able to nucleate microbubbles, as evidenced by a temporary increase in the scattering signals. These waveforms were recorded and processed the same way as above. Our motivation for performing this nucleation experiment was to compare the nucleated inertial cavitation (IC) signals with signals from destroyed contrast microbubbles. We will show below that, in most cases, the process of bubble destruction also resulted in inertial cavitation, either from preexisting or nucleated microbubbles.

Data acquisition and processing

To detect the subtle changes of signals, we continuously recorded 862 20 μs -long waveforms using the sequence mode of the oscilloscope. The detected signals were digitized by the oscilloscope at a resolution of 40 ns. The recorded waveforms were then processed by the Matlab™ program (Math Works, Natick, MA) on a PC to plot the waveforms and calculate the amplitude changes.

Various scattering patterns for BiSphere™ and Optison™ contrast microbubbles were selected from the experimental results and compared to the model obtained from the equations above. The initial radius R_0 , size distribution $W(R_0)$, SD of the distribution, and total relative number of bubbles were used as fitting parameters in the model.

RESULTS

Simulation

Bubble dissolution. Dissolution curves for two different sized air bubbles (1.5 and 2.5 μm) using eqn (1)

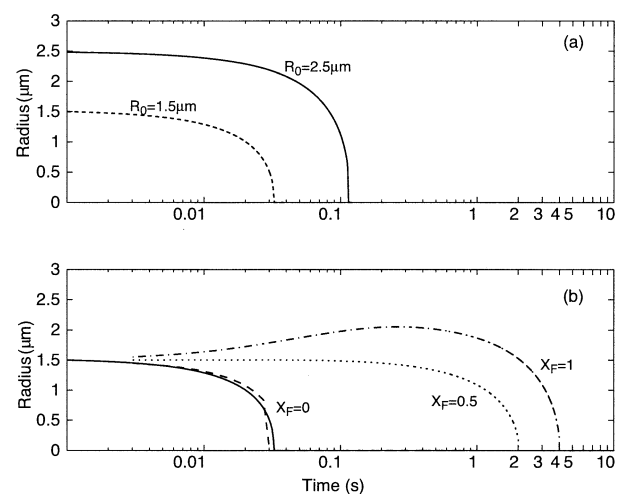


Fig. 4. (a) Predicted dissolution curves for 1.5- and 2.5- μm radius air bubbles calculated using eqn (1). (b) Dissolution curves for a 1.5- μm radius bubble with various C_3F_8 concentrations, using the mixed-gas equations. $X_F = 1$ implies that the bubble initially contains only OFP in air-saturated water, and $X_F = 0$ refers to an air bubble. For comparison with $X_F = 0$ is the dissolution curve of an air bubble calculated with eqn (1) (solid line).

are shown in Fig. 4a. It is appropriate to use eqn (1) for unshelled BiSphere™ agents because they contain air. However, Optison™ contrast bubbles contain OFP (C_3F_8), a much less soluble gas than air, which is not dissolved in the surrounding solution before the addition of the contrast agent. Thus, we used eqns (2), (3) and (5) to model the dissolution of this agent. As shown in the various dashed curves in Fig. 4b, after breaking the shell by high ultrasonic pressure, the free OFP bubble will experience diffusion of both air and OFP. The rate of diffusion will depend on the initial bubble contents and size, the dissolved gas in the fluid, and on the diffusivity of the gases involved. For high concentrations of OFP relative to air within the bubble (X_F greater than about 0.5), initial inward diffusion of air will cause the bubble to grow temporarily. Then, after the concentration balance of air is achieved (or when surface tension dominates), the bubble will dissolve. A comparison of eqn (1) with eqns (2), (3), and (5) for $X_F = 0$ (no C_3F_8) is also shown in Fig. 4b.

Scattering cross-section. Before we can implement the scattering cross-section equation, we must determine the frequency response of our system $H(f)$ to account for the transducer characteristics. We used a 10-cycle tone burst with a steel plate calibration reflector set to 45° to our bistatic configuration to record the system response. The frequency response fit well with a narrow-band Gaussian function with center frequency 5 MHz, and

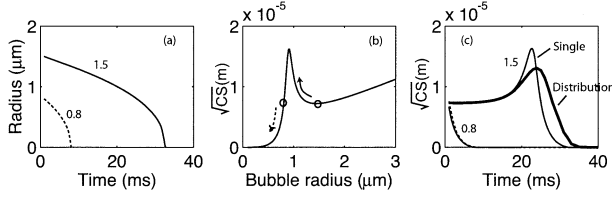


Fig. 5. (a) Radius vs. time curves for a 1.5- and 0.8- μm air bubble. (b) Scattering cross-section vs. radius curve. Note that a 1.5- μm bubble should experience an increased scattering cross-section because it dissolves through resonance. (c) Scattering cross-section vs. time for a 1.5- and 0.8- μm radius bubble. The small bubble decays as $1/R^6$, and the larger bubble undergoes resonance enhancement. Also shown is the effect of including a small Gaussian size distribution. The distribution lowers and broadens the resonance peak. $\sqrt{\text{CS}}$ = square root of scattering cross-section.

SD = 0.4 MHz, evaluated between 3 and 7 MHz. Thus, the scattering cross-section for the bubble distribution is given by:

$$\sigma_f = W(R_0)\sigma_s H(f). \quad (10)$$

The scattering cross-section from a 1.5 ± 0.03 and 0.8 ± 0.03 μm radius size distribution of free air bubbles in water is shown in Fig. 5. The resonance peak for a single air bubble at 5 MHz is approximately 0.9 μm . As the larger bubble (1.5 μm) dissolves, its scattering strength initially decreases before increasing substantially as it passes through resonance. However, the smaller bubble (0.8 μm) is initially smaller than resonance and, thus, its scattering signal decreases as the sixth power of its size. Inclusion of a bubble size distribution broadens out the resonance peak.

Experiments

In Fig. 6, is an experimental recording of the scattering signal from biSphere™ contrast bubbles. A steady high baseline signal is indicative of bubbles being in the focus of the HIFU bistatic detector system. After the first HIFU pulse (indicated above the scattering signal), the signal immediately jumps to a high level, followed by a quickly decreasing amplitude (labeled A in the plot) back to near the baseline value. The baseline signal slowly decreases for approximately 20 ms before undergoing a secondary increase, followed by a decay to the noise background level. With subsequent HIFU pulses, we observe only a small momentary increase in the scattering signal, possibly due to nucleation of small microbubbles.

To understand the scattering patterns, we performed simulations of the scattering amplitude based on the models mentioned previously to fit the experimental

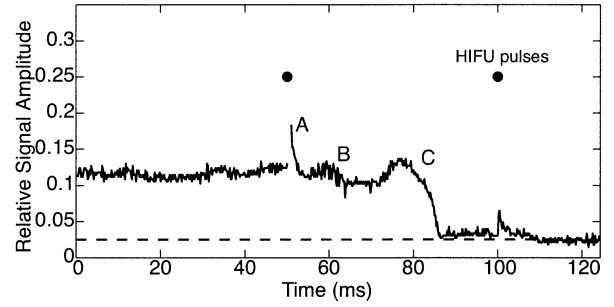


Fig. 6. The scattered signals from biSphere™ bubbles before and after the destruction process. Before the HIFU pulse, there is a high baseline signal indicating the presence of bubbles. Immediately after the HIFU pulse, there is a fast-decaying signal (A) back to baseline, followed by (B) a slowly decreasing signal, (C) a resonance-like feature, and a secondary decay to background levels. Subsequent HIFU pulses show only small decay signals, probably due to nucleation of microbubbles.

data. We found that no single bubble size distribution could account for all the features we observed. We, thus, developed a multimodal size distribution hypothesis that assumes the total signal is due to scattering from multiple bubble size distributions. For example, in the data set of Fig. 6, we assumed that each major feature (labeled A, B, and C) was associated with independent scattering from a particular bubble size distribution. The bubble size distribution for each contributor is shown in Fig. 7 (labeled 1a, 2a, and 3a). The associated scattering amplitude from each bubble size distribution is also shown in Fig. 7 (labeled 1b, 2b, and 3b).

We sum up the individual contributions, as well (Fig. 7, 4a and 4b). For the simulation to match the data, we must assume a relative population size for each contributed bubble size distribution. The relative sizes are, thus, also shown. For this data set, there are apparently many more small bubbles generated, probably due to nucleated microbubbles.

We also performed experiments in the absence of shelled microbubbles. At sufficiently high pressure amplitude, we were able to nucleate microbubbles and record their decay using acoustic scattering. Figure 8 shows three typical curves that represent various decay characteristics. In Fig. 8a, we observe a fast decay signal that corresponds to nucleated microbubbles smaller than resonance size. Because of their size, the decay curve follows a $1/R^6$ profile. In Fig. 8b, is a decay pattern that comes from a bubble size distribution larger than resonance. Thus, as the bubbles decay, they pass through resonance, exhibiting a momentary increase in scattering. In Fig. 8c, we show a bimodal distribution of bubbles. Each distribution exhibits a characteristic decay rate. Fitting of each of these curves shows good agreement with the data.

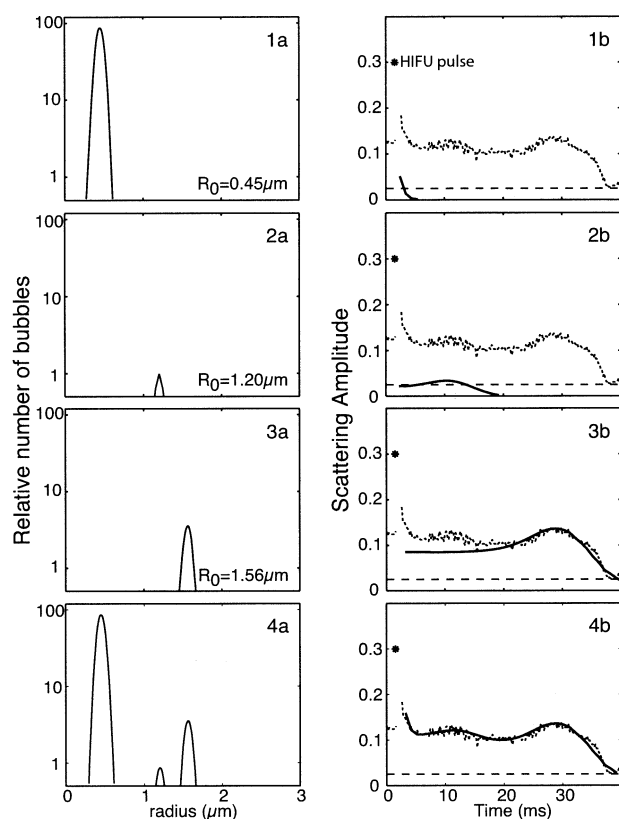


Fig. 7. The fast decay signal (1b) immediately after the HIFU application (labeled A in Fig. 6) is modeled as a distribution of nucleated microbubbles with $R_0 \approx 0.45 \mu\text{m}$ (1a). After the fast decay (labeled B in Fig. 6), a small resonance feature (2b) is modeled with a bubble size distribution of $R_0 \approx 1.2 \mu\text{m}$ (2a). The major resonance feature (labeled C in Fig. 6) is modeled with a size distribution centered around $R_0 \approx 1.56 \mu\text{m}$ (3a). Finally, an independent addition of scattering signals from the various size distributions (4a) generates the curve shown in (4b).

The scattering signals from Optison™ bubbles were also traced. Because the dissolving rate of the internal gas, OFP, is much slower than that of air, an extending time scale with wider intervals was used. However, we were unable to observe OFP bubble dissolution through resonance size. The long dissolution time of OFP was a factor because, during these long periods, bubbles can easily move into or through the focal zone. We estimate that a bubble velocity of about 4 mm/s was sufficient for the bubble to traverse the focal zone completely in the time frame we used. Our data showed many signs of bubble motion (undulations in scattering signal amplitude) that could account for our inability to observe dissolution through resonance.

We did, however, observe Optison™ fragmentation due to the HIFU pulse. Figure 9 shows a typical example from our data sets. An initial high baseline indicates that

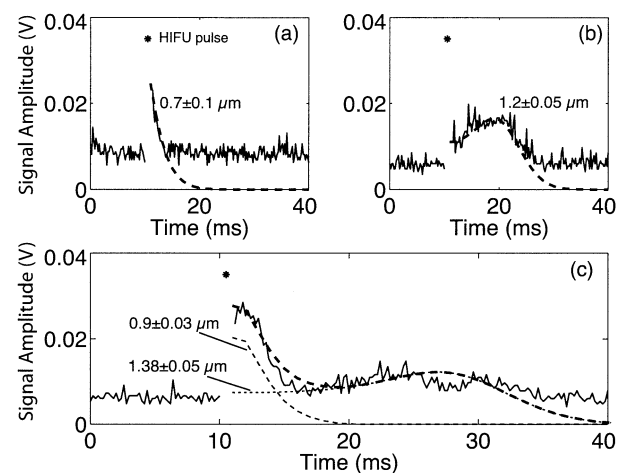


Fig. 8. The scattering signal of bubbles generated by inertial cavitation (nucleation). Three patterns were observed: (a) fast decay; (b) resonance-like bump; and (c) fast decay followed by a resonance-like bump. Based on the discussions related to Figs. 5 and 7, we conclude that it is probable that the bubble(s) in (a) are smaller than resonance, and the bubble(s) in (b) are slightly larger in resonance and decay through resonance, and that the bubbles in (c) formed a bimodal size distribution, with some smaller than resonance, and others slightly larger than resonance (with some effort, a trimodal size distribution can also be fit to the data). The fitted decay curves for each data set are shown as dashed lines. The combined distribution shown in (c) has a $\approx 2:1$ ratio of small to large bubbles.

Optison™ microbubbles are present at the focus. Immediately after the HIFU pulse, there is an increase in the scattering signal, followed by a fast decay to background levels. No resonant pattern was ever observed and, therefore, we believe that all the fragments from the destruction of Optison™ microbubbles are smaller than the resonance size.

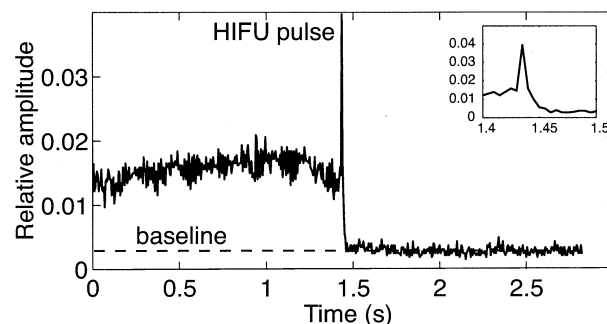


Fig. 9. The scattering signal from Optison™ contrast bubbles before and after destruction by a HIFU pulse. The pressure amplitude was 2.2 MPa. The decay rate was fitted to a bubble size distribution of $0.3 \pm 0.1 \mu\text{m}$ (not shown), suggesting that the bubbles fragmented. The inset shows the details of the scattering signal during bubble destruction.

DISCUSSION

We have shown the destruction process of US contrast agents using acoustic interrogation methods. In particular, we found that the destruction process can also be accompanied by nucleation. Although bubble destruction generates a range of bubble sizes, the model can reasonably fit the data by examining only the major features from the scattering signal. That is, the position and width of the resonance features provides a limit on available bubble sizes and distributions for inclusion into the model. The relative amplitude of the fast-decay signal gives us a hint as to the relative number of smaller bubbles. Using sizes, distributions and numbers outside these limits results in poor fits.

An interesting aspect of the resonance features is that it allows us to differentiate between bubbles of various sizes. The presence of a resonance depends on the initial bubble concentration and pulse parameters. Not all experiments generate resonance-like signals because sometimes the bubbles fragment to smaller-than-resonance size.

Contrast agent population and size distribution

We can estimate the number of bubbles in the focal zone of our system as follows. The concentration of Optison™ microbubbles is 5 to 8×10^8 spheres per mL, and approximately 5×10^8 microbubbles per mL for biSphere™ agents. We used $1 \mu\text{L}$ per L of water in our experiments. The -3-dB pressure contour for the 1.1-MHz HIFU transducer is cigar-shaped with a length of 10 mm and center diameter of 1 mm (the calculated focal volume is approximately 5 mm^3). Assuming that the microbubbles are evenly distributed throughout the volume of water, there are, thus, about two to three bubbles within the -3-dB focal volume at any given time.

However, our observations of shell-disrupted bubbles dissolving through resonance showed that the resonance peak is smaller and broader than one would observe from a single bubble. To fit the broadened resonant peak of the recorded signal, we assume that there is a normal distribution of bubble sizes with a small SD in radii. For the data in Fig. 7, the deviation was $\pm 0.05 \mu\text{m}$. This distribution effectively lowers and broadens the resonance peak. Note that this assumption is not necessarily entirely correct. It may also be that the bubbles are not entirely freed from the shell, which would also affect their dissolution curve, or that bubble motion through the focal zone affected the scattering signal. However, our results are consistent and plausible.

Nucleation and inertial cavitation

For nucleation, it is also reasonable to have a multiple-bubble assumption. Photographs taken during cav-

itation nucleation indicate that numerous small bubbles can be generated (Crum and Nordling 1972). This process might repeat several times within a HIFU tone burst, and generate many nuclei. The question remains, are the fast decay signals due to nucleation or bubble fragmentation?

The answer depends on the pressure amplitude, as well as the scattering signal amplitude. For instance, the data in Fig. 7 clearly show that following the HIFU pulse and initial fast decay, the scattering signal returns to the same level it had before the HIFU pulse. One would expect that, if the bubbles were fragmented, the signal after the HIFU pulse would be lower than that before the HIFU pulse. Because it is about the same level as before the pulse, we assume that the fast decay signal is due to nucleation of microbubbles, not fragmentation of pre-existing bubbles.

On the other hand, the Optison™ data in Fig. 9 shows that, following the HIFU pulse, the signal decays quickly to background, suggesting that these bubbles have been fragmented. For this particular data set, we found that an average fragment size of about $0.3 \mu\text{m}$ can best fit the decay signal. This is consistent with previous research (Dayton *et al.* 1999). Shell disruption without fragmentation would have led to much longer dissolution times.

Finally, by performing experiments in water, we were able to show various decay characteristics from nucleated microbubbles. Figure 8 shows that bubbles can be nucleated with a wide distribution of sizes. We found nucleated microbubbles below and above the resonance size (at 5 MHz).

Bubble translation

We examined bubble translation induced by the acoustic radiation force through the focal region to determine if such motion could account for the loss of scattering signals we observed. Bubble translation manifests itself as a change in the received echo time. To the resolution of our system, we were unable to observe bubble motion from the low-pressure output of the 5-MHz interrogating transducer (0.09 MPa peak amplitude at the focus). However, bubble motion was observed for HIFU pulses as short as five cycles, provided the pressure was high enough. The peak focal pressure of the HIFU transducer in our experiments ranged from 0.5 to 2 MPa . No significant change was found for low pressure levels; although, for the high-pressure conditions, a $< 1\text{-}\mu\text{s}$ shift was observed. After the HIFU pulse was terminated, the bubble motion came to a rest within a few recorded segments. Thus, the scattering amplitude change we observed was due to the bubble dissolution, not bubble motion.

In conclusion, we have observed US contrast mi-

crobubble dissolution after shell disruption. Air-filled agents such as biSpheres™ generated good decay characteristics that can be modeled relatively well. Our results indicate that, often, shell disruption is accompanied with inertial cavitation nucleation (at least, in water). We were able to fit the decay curves using single and multimodal bubble distributions very effectively. The relatively fast decay times for Optison™ bubbles suggest that these bubbles are often fragmented before undergoing dissolution. The implication is important to researchers using perfluorocarbon bubbles for reperfusion studies. If such bubbles did not fragment, one would expect to see the free gas lasting for many s, contaminating reperfusion estimates.

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REFERENCES

- Blomley MJ, Albrecht T, Cosgrove DO, et al. Stimulated acoustic emission to image a late liver and spleen-specific phase of Levovist in normal volunteers and patients with and without liver disease. *Ultrasound Med Biol* 1999;25:1341–1352.
- Bouakaz A, Frinking PJ, de Jong N, Bom N. Noninvasive measurement of the hydrostatic pressure in a fluid-filled cavity based on the disappearance time of micrometer-sized free gas bubbles. *Ultrasound Med Biol* 1999;25:1407–1415.
- Bouchot C, Richon D. Gas, liquid, and saturated vapor-liquid densities and vapor pressures of difluoromethane (R-32) and octafluoropropane (R-218) at temperatures from 253 K to 333 K and pressure up to 15.5 MPa. *Int Electron J Phys Chem* 1997;3:1–16.
- Chomas JE, Dayton P, Allen J, Morgan K, Ferrara KW. Mechanisms of contrast agent destruction. *IEEE Trans Ultrason Ferroelect Freq Control* 2001;48:232–248.
- Church CC. The effects of an elastic solid surface layer on the radial pulsations of gas bubbles. *J Acoust Soc Am* 1995;97:1510–1521.
- Crum LA, Nordling DA. Velocity of transient cavities in an acoustic stationary wave. *J Acoust Soc Am* 1972;52:294–301.
- Dayton PA, Morgan KE, Klibanov AL, Brandenburger G, Ferrara KW. Optical and acoustical observation of the effects of ultrasound on contrast agents. *IEEE Trans Ultrason Ferroelect Freq Control* 1999;46:220–232.
- Epstein PS, Plesset MS. On the stability of gas bubbles in liquid-gas solutions. *J Chem Physics* 1950;18:1505–1509.
- Everbach EC, Francis CW. Cavitation mechanisms in ultrasound-accelerated thrombolysis at 1 MHz. *Ultrasound Med Biol* 2000;26:1153–1160.
- Frinking PJA, Cespedes EI, Kirkhorn J, Torp HG, de Jong N. A new ultrasound contrast imaging approach based on the combination of multiple imaging pulses and a separate release burst. *IEEE Trans Ultrason Ferroelect Freq Control* 2001;48:643–651.
- Goldberg BB, ed. *Ultrasound contrast agents*. London: Martin Dunitz, 1997.
- Hayduk W, Laudie H. Prediction of diffusion coefficients for nonelectrolytes in dilute aqueous solutions. *AIChE J* 1974;20:611–615.
- Hoff L. Acoustic characterization of contrast agents for medical ultrasound imaging. Ph.D. thesis, Norwegian University of Science and Technology, Trondheim, Norway, 2000.
- Hoff L, Sontum PC, Hovem JM. Oscillations of polymeric microbubbles: Effect of the encapsulating shell. *J Acoust Soc Am* 2000;107:2272–2280.
- Kabalinov A, Klein D, Pelura T, Schutt E, Weers J. Dissolution of multicomponent microbubbles in the bloodstream: 1. Theory. *Ultrasound Med Biol* 1998;24:739–749.
- Kabalinov A, Makarov KN, Shcherbakova OV. Solubility of fluorocarbons in water as a key parameter determining fluorocarbon emulsion stability. *J Fluor Chem* 1990;50:271–284.
- Kamiyama N, Moriyasu F, Mine Y, Goto Y. Analysis of flash echo from contrast agent for designing optimal ultrasound diagnostic systems. *Ultrasound Med Biol* 1999;25:411–420.
- Lawson DD, Moacanin J, Scherer JKV, Terranova TF, Ingham JD. Methods for the estimation of vapor pressures and oxygen solubilities of fluorocarbons for possible application in artificial blood formulations. *J Fluor Chem* 1978;12:221–236.
- Lombardo A, Loperfido F. Myocardial contrast echocardiography: A new tool in the noninvasive assessment of perfusion. *Rays* 2001;26:45–51.
- Nishioka T, Luo H, Fishbein MC, et al. Dissolution of thrombotic arterial occlusion by high intensity, low frequency ultrasound and dodecafluoropentane emulsion: An in vitro and in vivo study. *J Am Coll Cardiol* 1997;30:561–568.
- Pelberg RA, Wei K, Kamiyama N, et al. Potential advantage of flash echocardiography for digital subtraction of B-mode images acquired during myocardial contrast echocardiography. *J Am Soc Echo Cardiogr* 1999;12:85–93.
- Podell S, Burrascano C, Gaal M, et al. Physical and biochemical stability of Optison, an injectable ultrasound contrast agent. *Bio-technol Appl Biochem* 1999;30:213–223.
- Pohl C, Tiemann K, Schlosser T, Becher H. Stimulated acoustic emission detected by transcranial color Doppler ultrasound: A contrast-specific phenomenon useful for the detection of cerebral tissue perfusion. *Stroke* 2000;31:1661–1666.
- Porter TR, Everbach C, Kricsfeld D, Xie F. Myocardial cavitation activity during continuous infusion and bolus intravenous injections of perfluorocarbon-containing microbubbles. *J Am Soc Echocardiogr* 2001;14:618–625.
- Porter TR, Iversen PL, Li S, Xie F. Interaction of diagnostic ultrasound with synthetic oligonucleotide-labeled perfluorocarbon-exposed sonicated dextrose albumin microbubbles. *J Ultrason Med* 1996;15:577–584.
- Schlosser T, Pohl C, Veltmann C, et al. Feasibility of the flash-replenishment concept in renal tissue: Which parameters affect the assessment of the contrast replenishment? *Ultrasound Med Biol* 2001;27:937–944.
- Shi WT, Forsberg F, Tornes A, Ostensen J, Goldberg BB. Destruction of contrast microbubbles and the association with inertial cavitation. *Ultrasound Med Biol* 2000;26:1009–1019.
- Shohet RV, Chen S, Zhou YT, et al. Echocardiographic destruction of albumin microbubbles directs gene delivery to the myocardium. *Circulation* 2000;101:2554–2556.
- Sieswerda GT, Kamp O, Visser CA. Myocardial contrast echocardiography: Clinical benefit and practical issues. *Echocardiography* 2000;17:S25–S36.
- Tiemann K, Pohl C, Schlosser T, et al. Stimulated acoustic emission: Pseudo-Doppler shifts seen during the destruction of nonmoving microbubbles. *Ultrasound Med Biol* 2000;26:1161–1167.
- Vaezy S, Martin R, Crum L. High intensity focused ultrasound: A method of hemostasis. *Echocardiography* 2001;18:309–315.
- Wei K, Jayaweera AR, Firoozan S, et al. Quantification of myocardial blood flow with ultrasound-induced destruction of microbubbles administered as a constant venous infusion. *Circulation* 1998;97:473–483.

APPENDIX A

Estimation of the diffusion and OSTWALD constants for OFP

The diffusion coefficient D_F of OFP in 25°C water is not readily available in the literature and, thus, was evaluated with the empirical equation from Hayduk and Laudie (1974), which includes the estimation of the molecular volume (V_m , cm³/mol) of OFP:

$$D_F = 13.26 \times 10^{-9} \cdot V_m^{-0.589}. \quad (A1)$$

The molecular volume was calculated by the group additivity method, which considered the contributions of the two CF_3 - groups and one $-\text{CF}_2-$ group (Lawson et al. 1978). The estimated molecular volume was $132.7 \text{ cm}^3/\text{mol}$, and the calculated diffusion coefficient was, thus, $7.45 \times 10^{-10} \text{ m}^2/\text{s}$, which lies between the diffusion coefficients of C_2F_6 and C_4F_{10} in Kabalnov et al. (1990), and is close to the value obtained by another method, the Stokes-Einstein approximation ($7.72 \times 10^{-10} \text{ m}^2/\text{s}$) (Chomas et al. 2001).

The Ostwald coefficient is defined as the dimensionless ratio of the solubility of the gas in the liquid to the gas density, eqn (A2). For nitrogen and oxygen, the coefficients are available in the literature (Kabalnov et al. 1998). Thus, the Ostwald coefficient of air can be calculated by volume averaging of the nitrogen and oxygen. However, the Ostwald coefficient for OFP (L_F) is not available and needs to be calculated. In the homologous series of nonbranched fluorocarbons, the solubility (molar concentration of gas in solution) is lowered by a factor of approximately 8 for each additional $-\text{CF}_2-$ group (Kabalnov et al. 1990). The water solubility of

C_2F_6 is 1.45 mol/m^3 and, therefore, the solubility of OFP was calculated as 0.181 mol/m^3 . The saturated vapor concentration of OFP can be evaluated by the ideal gas equation, eqn (A3), and the vapor pressure of OFP can be calculated from the equation, eqn (A4), provided by Bouchot and Richon (1997), which was based on measurements using the vibrating tube densimeter method. The calculated vapor pressure of OFP is $8.64 \times 10^5 \text{ Pa}$ and the Ostwald coefficient is, thus, 5.20×10^{-4} . These two values, again, fall between the values of C_2F_6 and C_4F_{10} in the literature (Kabalnov et al. 1998).

$$L = \frac{c_0}{\rho_g} \quad (\text{A2})$$

$$c_0 = \frac{MP_v}{R_g T} \quad (\text{A3})$$

$$P_v = \exp(A + B/T), \quad (\text{A4})$$

where $A = 21.784 \pm 0.014$ and $B = -2418.1 \pm 4.6$.